

Relatório Laboratório 2 C-Academy Formação Avançada em Cibersegurança

Por: Joel Barrenho, Rui Carapinha, Samuel Bárias

Link Repo: <https://github.com/RuiCarapinha/Lab2>

1. a) ... d)

```
Microsoft Windows [Version 10.0.26100.7840]
(c) Microsoft Corporation. Todos os direitos reservados.

C:\Users\Admin>cd C:\Users\Admin\py\PTC-lab1\lab2-alunos

C:\Users\Admin\py\PTC-lab1\lab2-alunos>C:\Users\Admin\AppData\Local\Microsoft\WindowsApps/python3.11.exe c:/Users/Admin/py/PTC-lab1/lab2-alunos/client_todo.py
** Todo List Menu **
1. Add todo
2. Print todos
3. Complete todo
10. Exit
Enter your choice:
```

```
Microsoft Windows [Version 10.0.26100.7840]
(c) Microsoft Corporation. Todos os direitos reservados.

C:\Users\Admin>cd C:\Users\Admin\py\PTC-lab1\lab2-alunos

C:\Users\Admin\py\PTC-lab1\lab2-alunos>c:/Users/Admin/py/PTC-lab1/lab2-alunos/server_todo.py
C:\Users\Admin\py\PTC-lab1\lab2-alunos>& C:/Users/Admin/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/Admin/py/PTC-lab1/lab2-alunos/server_todo.py
& was unexpected at this time.

C:\Users\Admin\py\PTC-lab1\lab2-alunos>& C:/Users/Admin/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/Admin/py/PTC-lab1/lab2-alunos/server_todo.py
& was unexpected at this time.

C:\Users\Admin\py\PTC-lab1\lab2-alunos>& C:/Users/Admin/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/Admin/py/PTC-lab1/lab2-alunos/server_todo.py
& was unexpected at this time.

C:\Users\Admin\py\PTC-lab1\lab2-alunos>C:/Users/Admin/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/Admin/py/PTC-lab1/lab2-alunos/server_todo.py
Waiting for connection...
```

```
Enter your choice: 2
Print todos
Connected to server.
Connection close.

** Todo List Menu **
1. Add todo
2. Print todos
3. Complete todo
10. Exit
Enter your choice: 1
Invalid choice
** Todo List Menu **
1. Add todo
2. Print todos
3. Complete todo
10. Exit
Enter your choice: 1
Enter todo title: a1
Enter todo description: b1
Connected to server.
Connection close.
Todo added.

** Todo List Menu **
1. Add todo
2. Print todos
3. Complete todo
10. Exit
Enter your choice: 2
Print todos
Connected to server.
Connection close.
0. [ ] a1: b1

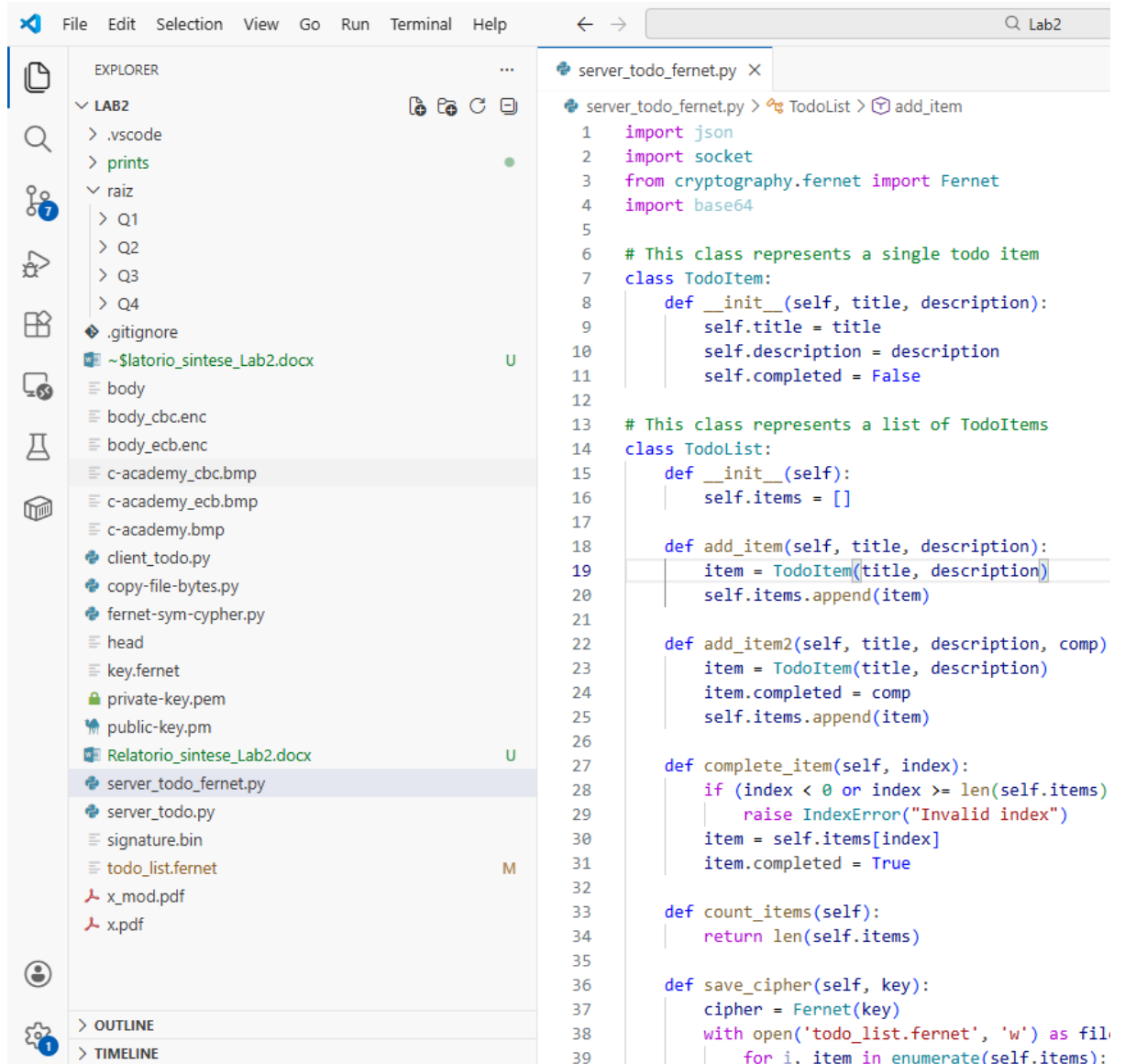
** Todo List Menu **
1. Add todo
2. Print todos
3. Complete todo
10. Exit
Enter your choice: 1
Enter todo title: so um teste
Enter todo description: descrição
Connected to server.
Connection close.
Todo added.

** Todo List Menu **
1. Add todo
2. Print todos
3. Complete todo
10. Exit
Enter your choice:
```

```
Traceback (most recent call last):
  File "C:\Users\Admin\py\PTC-lab1\lab2-alunos\server_todo.py", line 51, in <module>
    client_socket, address = server_socket.accept()
    ^^^^^^^^^^^^^^^^^^^^^
KeyboardInterrupt
C:\Users\Admin\py\PTC-lab1\lab2-alunos>C:/Users/Admin/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/Admin/py/PTC-lab1/lab2-alunos/server_todo.py
Waiting for connection...
Connected to ('127.0.0.1', 60928)
Received command: 2-
Logging:
Waiting for connection...
Connected to ('127.0.0.1', 60929)
Received command: 2-
Logging:
Waiting for connection...
Traceback (most recent call last):
  File "C:\Users\Admin\py\PTC-lab1\lab2-alunos\server_todo.py", line 51, in <module>
    client_socket, address = server_socket.accept()
    ^^^^^^^^^^^^^^^^^^^^^
KeyboardInterrupt
C:\Users\Admin\py\PTC-lab1\lab2-alunos>C:/Users/Admin/AppData/Local/Microsoft/WindowsApps/python3.11.exe c:/Users/Admin/py/PTC-lab1/lab2-alunos/server_todo.py
Waiting for connection...
Connected to ('127.0.0.1', 50782)
Received command: 2-
Logging:
Waiting for connection...
Connected to ('127.0.0.1', 60175)
Received command: 2-
Logging:
Waiting for connection...
Connected to ('127.0.0.1', 60170)
Received command: 1-a1,b1
Logging: Todo added.
Waiting for connection...
Connected to ('127.0.0.1', 60183)
Received command: 2-
Logging: 0. [ ] a1: b1

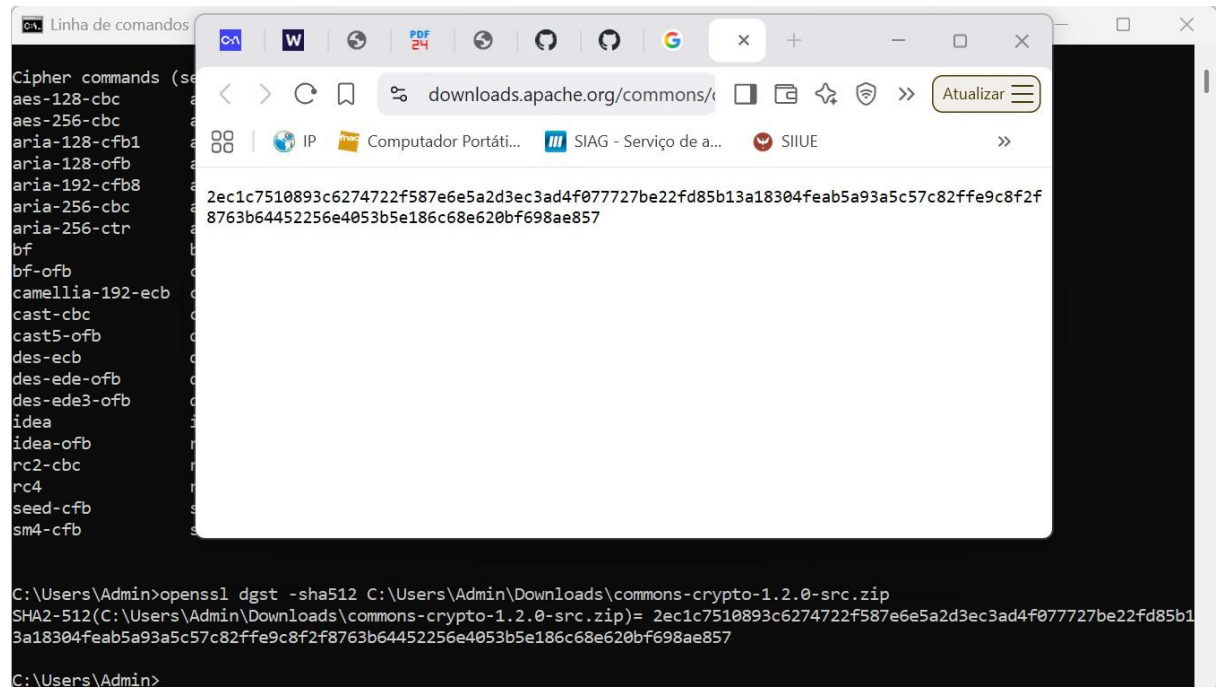
Waiting for connection...
Connected to ('127.0.0.1', 60186)
Received command: 1-so um teste,descrição
Logging: Todo added.
Waiting for connection...
```

2. a) ... g)



```
File Edit Selection View Go Run Terminal Help
server_todo_fernet.py X
server_todo_fernet.py > TodoList > add_item
1 import json
2 import socket
3 from cryptography.fernet import Fernet
4 import base64
5
6 # This class represents a single todo item
7 class TodoItem:
8     def __init__(self, title, description):
9         self.title = title
10        self.description = description
11        self.completed = False
12
13 # This class represents a list of TodoItems
14 class TodoList:
15     def __init__(self):
16         self.items = []
17
18     def add_item(self, title, description):
19         item = TodoItem(title, description)
20         self.items.append(item)
21
22     def add_item2(self, title, description, comp):
23         item = TodoItem(title, description)
24         item.completed = comp
25         self.items.append(item)
26
27     def complete_item(self, index):
28         if (index < 0 or index >= len(self.items)):
29             raise IndexError("Invalid index")
30         item = self.items[index]
31         item.completed = True
32
33     def count_items(self):
34         return len(self.items)
35
36     def save_cipher(self, key):
37         cipher = Fernet(key)
38         with open('todo_list.fernet', 'w') as file:
39             for i, item in enumerate(self.items):
```

3.
a).
i. e ii.



C:\Users\Admin\Downloads commons-crypto-1.2.0-src.zip - Notepad++

Ficheiro Editar Procurar Visualização Codificação Linguagem Definições Ferramentas Macro Executar Plugins Janela ?

commons-crypto-1.2.0-src.zip

1 ePKETXEOT

2 NULNULBSNULNUL>¿UNULNULNULNULNULNULNULNULNULNULEMNULNULNUL
commons-crypto-1.2.0-src/PKETXEOT

3 NULNULBSNULNUL>¿UNULNULNULNULNULNULNULNULNULNULGSNULNULNUL
commons-crypto-1.2.0-src/src/PKETXEOT

4 NULNULBSNULNUL>¿UNULNULNULNULNULNULNULNULNUL&NULNULNUL
commons-crypto-1.2.0-src/src/assembly/PKETXEOT

5 NULNULBSNULNUL>¿UNULNULNULNULNULNULNULNULNUL%NULNULNUL
commons-crypto-1.2.0-src/src/changes/PKETXEOT

6 NULNULBSNULNUL>¿UNULNULNULNULNULNULNULNULNUL\$NULNULNUL
commons-crypto-1.2.0-src/src/docker/PKETXEOT

7 NULNULBSNULNUL>¿UNULNULNULNULNULNULNULNULNUL"NULNULNUL
commons-crypto-1.2.0-src/src/main/PKETXEOT

8 NULNULBSNULNUL>¿UNULNULNULNULNULNULNULNULNUL'NULNULNUL
commons-crypto-1.2.0-src/src/main/java/PKETXEOT

9 NULNULBSNULNUL>¿UNULNULNULNULNULNULNULNULNUL+NULNULNUL
commons-crypto-1.2.0-src/src/main/java/org/PKETXEOT

10 NULNULBSNULNUL>¿UNULNULNULNULNULNULNULNULNUL2NULNULNUL
commons-crypto-1.2.0-src/src/main/java/org/apache/PKETXEOT

11 NULNULBSNULNUL>¿UNULNULNULNULNULNULNULNULNUL:NULNULNUL
commons-crypto-1.2.0-src/src/main/java/org/apache/commons/PKETXEOT

length : 282 109 lines : 2 058 Ln : 2 Col : 1 Sel : 5 | 1 Unix (LF) ANSI IN

Linha de comandos

aria-128-cfb1	aria-128-cfb8	aria-128-ctr	aria-128-ecb
aria-128-ofb	aria-192-cbc	aria-192-cfb	aria-192-cfb1
aria-192-cfb8	aria-192-ctr	aria-192-ecb	aria-192-ofb
aria-256-cbc	aria-256-cfb	aria-256-cfb1	aria-256-cfb8
aria-256-ctr	aria-256-ecb	aria-256-ofb	base64
bf	bf-cbc	bf-cfb	bf-ecb
bf-ofb	camellia-128-cbc	camellia-128-ecb	camellia-192-cbc
camellia-192-ecb	camellia-256-cbc	camellia-256-ecb	cast
cast-cbc	cast5-cbc	cast5-cfb	cast5-ecb
cast5-ofb	des	des-cbc	des-cfb
des-ecb	des-edc	des-edc-cbc	des-edc-cfb
des-edc-ofb	des-edc3	des-edc3-cbc	des-edc3-cfb
des-edc3-ofb	des-ofb	des3	desx
idea	idea-cbc	idea-cfb	idea-ecb
idea-ofb	rc2	rc2-40-cbc	rc2-64-cbc
rc2-cbc	rc2-cfb	rc2-ecb	rc2-ofb
rc4	rc4-40	seed	seed-cbc
seed-cfb	seed-ecb	seed-ofb	sm4-cbc
sm4-cfb	sm4-ctr	sm4-ecb	sm4-ofb

C:\Users\Admin>openssl dgst -sha512 C:\Users\Admin\Downloads commons-crypto-1.2.0-src.zip
SHA2-512(C:\Users\Admin\Downloads commons-crypto-1.2.0-src.zip)= 2ec1c7510893c6274722f587e6e5a2d3ec3ad4f077727be22fd85b1
3a18304feab5a93a5c57c82ffe9cf2f8763b64452256e4053b5e186c68e620bf698ae857

C:\Users\Admin>openssl dgst -sha512 C:\Users\Admin\Downloads commons-crypto-1.2.0-src.zip
SHA2-512(C:\Users\Admin\Downloads commons-crypto-1.2.0-src.zip)= 6e75d197c2be3861f423eeef6be546113453d945b25dbbd623ed9350
5341eb29e21361d5dd2f77e57ba10015b43197ef9ec01c42cd7be0e53bf61923bc0b7d4e

C:\Users\Admin>

3. b) i.

```

C:\Users\Admin>openssl dgst -sha512 C:\Users\Admin\Downloads\commons-crypto-1.2.0-src.zip
SHA2-512(C:\Users\Admin\Downloads\commons-crypto-1.2.0-src.zip)= 2ec1c7510893c6274722f587e6e5a2d3ec3ad4f077727b22fd85b1
8a18304feab5a93a5c57c82ffe9c8f2f8763b64452256e4053b5e186c68e620bf698ae857

C:\Users\Admin>openssl dgst -sha512 C:\Users\Admin\Downloads\commons-crypto-1.2.0-src.zip
SHA2-512(C:\Users\Admin\Downloads\commons-crypto-1.2.0-src.zip)= 6e75d197c2be3861f423eef6be546113453d945b25dbbd623ed9350
5341eb29e21361d5fdd2f77e57ba10015b43197ef9ec01c42cd7be0e53bf61923bc0b7d4e

C:\Users\Admin>openssl dgst -sha512 -hmac "chave" C:\Users\Admin\Downloads\Lab2-enunciado.pdf
HMAC-SHA2-512(C:\Users\Admin\Downloads\Lab2-enunciado.pdf)= 5e13bf119995ff3d56ef712b022723642a5e0085423b746f429ce3dad73c
c5a29c0dfbd189846bebe3540d070f274c2b54fd80888aa5ddbca1a919b57d7acaef

C:\Users\Admin>openssl

```

3. c) i.

Joel encriptou com a password “chave”

```

C:\Users\Admin>openssl enc -aes-256-cbc -in C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2\prints\teste.pdf -out C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2\prints\teste.enc
enter AES-256-CBC encryption password:

Verifying - enter AES-256-CBC encryption password:

*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

C:\Users\Admin>openssl enc -aes-256-cbc -d -in C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2\prints\teste.enc -out C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2\prints\teste_d.pdf
enc: Extra (unknown) options: "aes-256-cbc" "in"
enc: Use -help for summary.

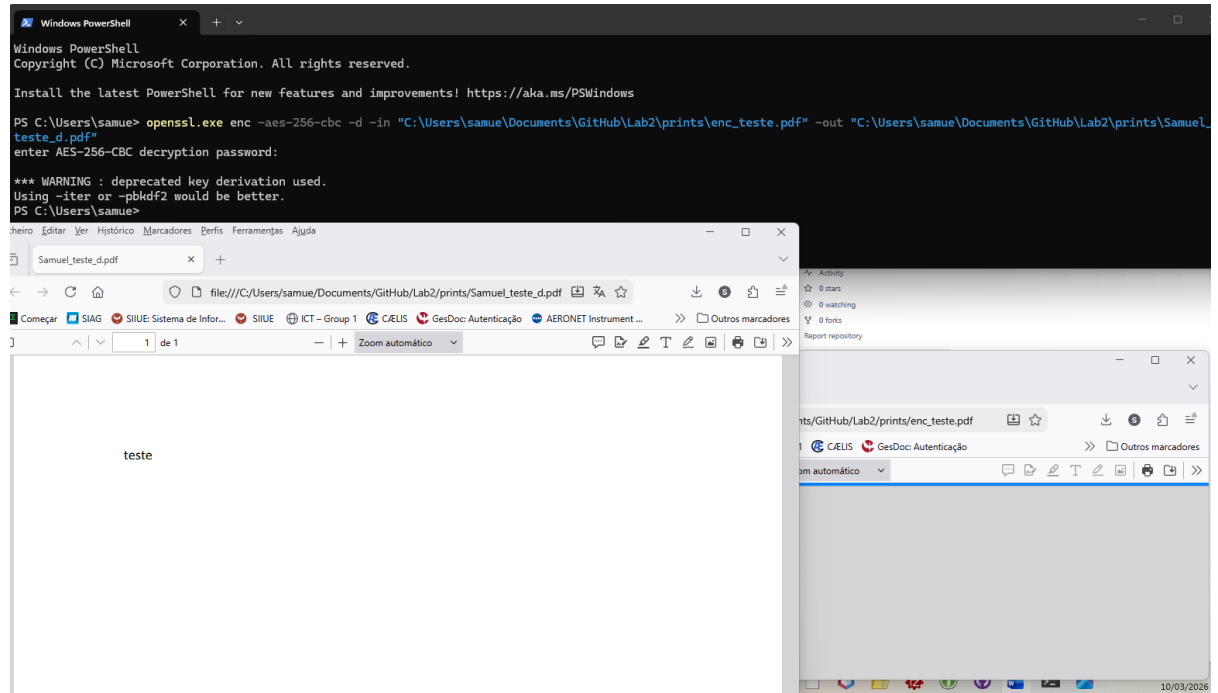
C:\Users\Admin>openssl enc -aes-256-cbc -d -in C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2\prints\teste.enc -out C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2\prints\teste_d.pdf
enter AES-256-CBC decryption password:

*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.

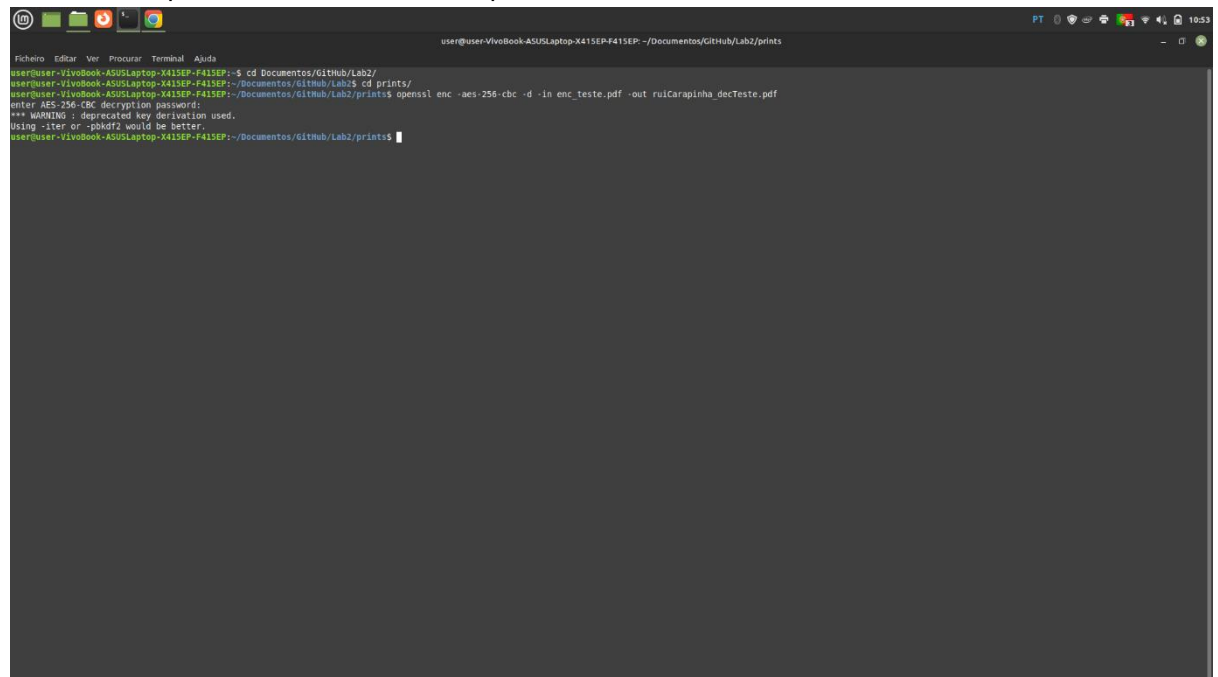
C:\Users\Admin>

```

Samuel descriptou com sucesso com a password “chave”



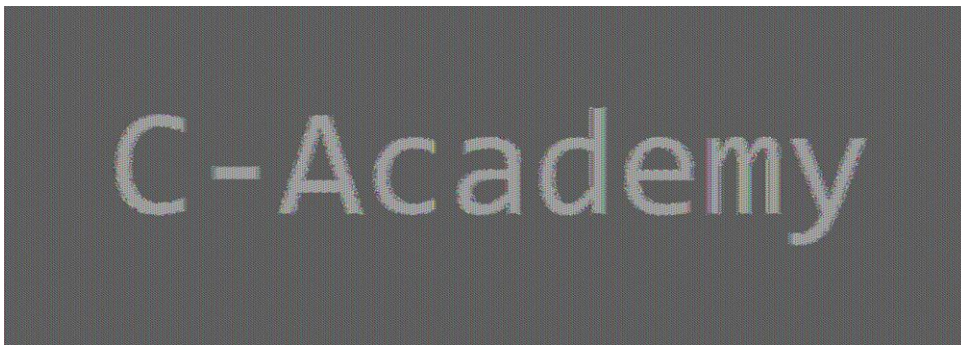
Rui descriptou com sucesso com a password “chave”



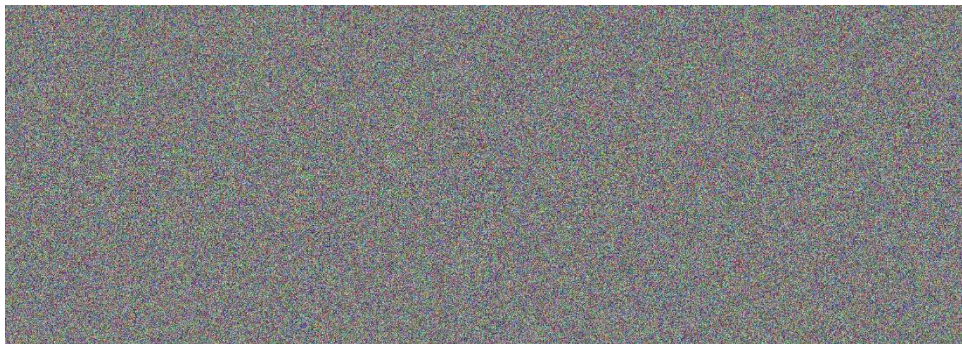
3.

C-Academy

d) i. ECB



d) ii. CBC



d) iv.

Através do modo ECB (Electronic Code book) como a informação é dividida em blocos(64-bits) e encriptada separadamente, podem-se identificar padrões, este exemplo mostra de forma clara que revelando padrões se pode ter indicações sobre a informação original.

Através do modo CBC (cipher-block chaining) como os blocos são ligados formando uma cadeia de blocos, os eventuais padrões são minimizados oferecendo um resultado mais eficaz.

3.

e)

```
C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2>openssl genrsa -out private-key.pem 2048

C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2>openssl dgst -sha256 -sign private-key.pem -out signature.bin x.pdf

C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2>openssl rsa -in private-key.pem -pubout -out public-key.pm
writing RSA key

C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2>openssl dgst -sha256 -verify public-key.pm -signature siganture.bin x.pdf
Error opening signature file siganture.bin
EC4D0000:error:80000002:system library:BIO_new_file:No such file or directory:crypto\bio\bss_file.c:67:calling fopen(siganture.bin, rb)
EC4D0000:error:10000080:BIO routines:BIO_new_file:no such file:crypto\bio\bss_file.c:75:

C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2>openssl dgst -sha256 -verify public-key.pm -signature signature.bin x.pdf
Verified OK

C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2>openssl dgst -sha256 -verify public-key.pm -signature signature.bin x_mod.pdf
Verification failure
082E0000:error:02000068:rsa routines:ossl_rsa_verify:bad signature:crypto\rsa\rsa_sign.c:441:
082E0000:error:1C800004:Provider routines:rsa_verify_directly:RSA lib:providers\implementations\signature\rsa_sig.c:1038:

C:\Users\Admin\py\PTC-lab1\lab2-alunos\github\Lab2>openssl dgst -sha256 -verify public-key.pm -signature signature.bin x_mod.pdf
```

4.

iii.

```
Windows PowerShell
PS C:\Users\samue\Documents\GitHub\Lab2> python.exe .\server_todo_fernet.py
Loaded key: b'CP8wKdabSKC7ldpr7465JWV2hrABZFD0H2Xbm6HnhRg='
going to decrypt: gAAAAABpqwHkwNzDGf7AYjETmNGhF0VaADFSWaSWMvzn-wAf8pc0Gz-soGEPIjddy44PxsNJHkVestc0eoL0s-0pe6Cw9VLL0g==
decrypted: Tarefa 4, False
Server started on port 24000...
Serving todo list...
0. [x] Tarefa 4:

Waiting for connection...
Connected to ('127.0.0.1', 50955)
Received command: 2-
Logging: 0. [x] Tarefa 4:

Waiting for connection...
Connected to ('127.0.0.1', 53818)
Received command: 1-Tarefa 1,Fazer tarefa 1
Logging: Todo added.
Waiting for connection...
Connected to ('127.0.0.1', 62262)
Received command: 2-
Logging: 0. [x] Tarefa 4:
1. [ ] Tarefa 1: Fazer tarefa 1

Waiting for connection...
Connected to ('127.0.0.1', 62270)
Received command: 1-Tarefa 2,Fazer tarefa 2
Logging: Todo added.
Waiting for connection...
Connected to ('127.0.0.1', 62271)
Received command: 3-2
Logging: Todo completed.
Waiting for connection...
Connected to ('127.0.0.1', 62272)
Received command: 2-
Logging: 0. [x] Tarefa 4:
1. [ ] Tarefa 1: Fazer tarefa 1
2. [x] Tarefa 2: Fazer tarefa 2

Waiting for connection...
|
```

iv.

```
Windows PowerShell X + v
3. Complete todo
4. nr itens para completar
10. Exit
Enter your choice: 1
Enter todo title: Tarefa 2
Enter todo description: Fazer tarefa 2
Connected to server.
Connection close.
Todo added.
** Todo List Menu **
1. Add todo
2. Print todos
3. Complete todo
4. nr itens para completar
10. Exit
Enter your choice: 3
Complete todo
Enter index: 2
Connected to server.
Connection close.
Todo completed.
** Todo List Menu **
1. Add todo
2. Print todos
3. Complete todo
4. nr itens para completar
10. Exit
Enter your choice: 2
Print todos
Connected to server.
Connection close.
0. [x] Tarefa 4:
1. [ ] Tarefa 1: Fazer tarefa 1
2. [x] Tarefa 2: Fazer tarefa 2

** Todo List Menu **
1. Add todo
2. Print todos
3. Complete todo
4. nr itens para completar
10. Exit
Enter your choice: |
```

```
Ficheiro Editor Ver Procurar Terminal Ajuda
user@user-VivoBook-ASUSLaptop-X415EP-F415EP: ~/Documents/GitHub/Lab2/
user@user-VivoBook-ASUSLaptop-X415EP-F415EP:~$ cd Documents/GitHub/Lab2/
user@user-VivoBook-ASUSLaptop-X415EP-F415EP:~/Documents/GitHub/Lab2$ cd prints/
user@user-VivoBook-ASUSLaptop-X415EP-F415EP:~/Documents/GitHub/Lab2/prints$ openssl enc -aes-256-cbc -d -in enc_teste.pdf -out ruiCarapinha_decTeste.pdf
enter AES-256-CBC decryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
user@user-VivoBook-ASUSLaptop-X415EP-F415EP:~/Documents/GitHub/Lab2/prints$
```

1

gAAAAABpr_EYkrQvjkit32L-nE20Cxvj0sfmqVwqvId--ZzdhJ8DkOw1OgrTkh9oFXnb8DNXUbm6v7bZLQwYKf_6uy2enOpqg==

2

gAAAAABpr_EYJ-yDTtoOKqLmmXx21Fq2-Sqe922hmn1_xBBpg37jsRTuIBtRHFcbvagmlqU66fBprVOV8C5aRrJnT8MDAqhPDyv_hqinmCK3FXDKr1iNNGtE=

3

gAAAAABpr_EYcnKhyUKnW3J4mu4hHuORU8kDRetWXQzeASavcn1HK1GxKub9zzVc2bf8v1MBsGG2MR0F1Y8te9W0Pk0_s1VXQJ7NO6kLYrMTen9HJv7WoVA=

4